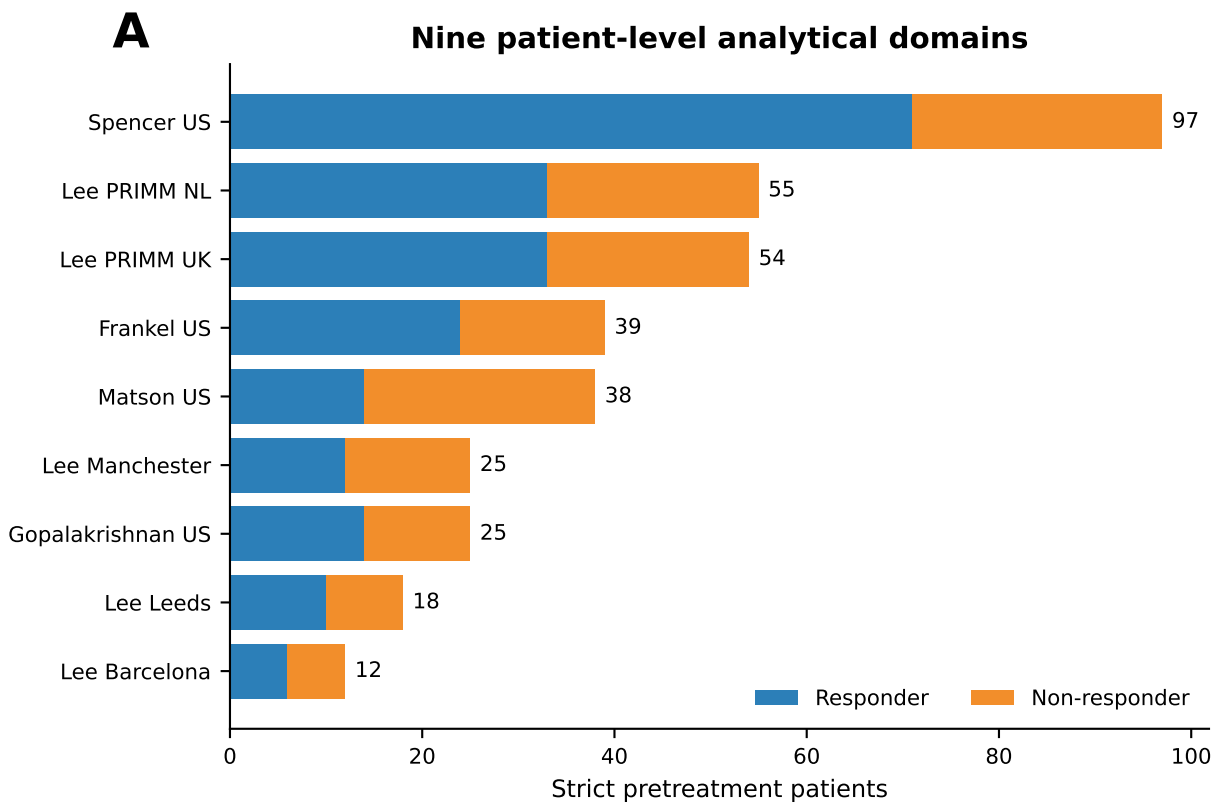


Patient-level harmonization, publication-family structure, and analytical design



B **Patient-specimen-run-matrix governance**

- Patient identity**

One canonical pretreatment profile per patient; duplicate specimens and runs audited.
- Profile linkage**

Exact patient-sample-run-matrix-column crosswalk; Spencer dual accessions resolved.
- Endpoint adjudication**

Matson, Manchester, Barcelona and BCN-12_1 labels audited; PRIMM0069 excluded because no processed profile was available.
- Frozen population**

363 patients: 217 responders and 146 non-responders; 2,205 true species features.

C **Mandatory endpoint sensitivity scenarios**

- PRIMARY**

Final adjudicated response lock; n = 363.
- S1**

Exclude BCN-12_1; n = 362.
- S2**

Classify BCN-12_1 as non-responder; n = 363.
- S3**

Exclude Manchester and Barcelona domains; n = 326.

Every predictive and species-effect analysis was rerun under all four scenarios.

D **Publication-family hierarchy**

- Lee 2022 family**

Five distinct clinical domains: PRIMM UK, PRIMM NL, Manchester, Leeds and Barcelona.
- Single-domain families**

Frankel 2017; Gopalakrishnan 2018; Matson 2018; Spencer 2021.

All nine sites remain separate clinical domains. The five Lee sites are jointly withheld during publication-family-aware validation.

E **Prespecified omics-layer analysis gate**

- V2 MetaPhlAn4 species/SGB**

PRIMARY: species-level transportability and effect meta-analysis.
- V2 unstratified HUMAnN3 MetaCyc pathways**

Not included in the primary analysis because exact units and patient-column identifiers were not fully verified.
- V1 unstratified HUMAnN community pathways**

SEPARATE SENSITIVITY: five-study functional analysis and species-pathway concordance; never concatenated with V2.
- Taxon-stratified pathways**

Not included in the primary analysis; only targeted exploratory interpretation was allowed.

V1 functional profiles were analysed separately and were never concatenated with V2 species profiles.